Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Free Body Diagrams — ANSWERS

The forces you will draw

Weight, W, is the pull of gravity on the object: straight down.

Normal contact force, F_N, is a surface pushing on the object, at right angles to the surface.

Tension, F_T, is a rope, string or cable pulling on the object, along the cable.

Friction, F_f, is a surface resisting sliding, against the motion.

Air resistance, or **drag**, F_d, is the air resisting motion, against the motion.

Driving force, or **thrust**, F_{drive}, is an engine, propeller or swimmer pushing forwards.

Lift, F_L, is the air pushing up on a wing.

Braking force, F_{brake}, is friction from the brakes, against the motion.

Resistive forces are friction and air resistance together: $F_f + F_d$.

On an AQA paper, label each arrow with the **name**: the name is what earns the mark. The symbol is shorthand, written beside it. W is AQA's own symbol, from $W = mg$.

The method

A free body diagram shows one object and every force acting on it, each drawn as an arrow.

1. Draw the object as a simple shape.
2. For **every** force acting *on* the object, draw an arrow from its centre, pointing the way the force acts.
3. Make the arrow's **length** show the force's size: equal forces, equal arrows.
4. Label each arrow with the force's **name**, from the list above, and its **size**.
5. Find the resultant along each line: up–down, then left–right.

Resultant **zero**: the velocity stays the same. Resultant **not zero**: the velocity changes, in the direction of the resultant.

Worked example — copy this shape

A box is pushed across a floor and speeds up. The push is 30 N to the right and friction is 10 N. The box's weight is 50 N.

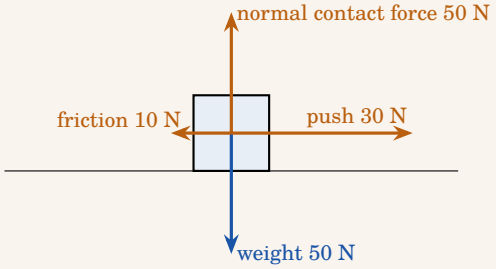
Up–down $50 - 50 = 0$

Left–right $30 - 10 = 20$ N to the right

Resultant **20 N to the right**

Velocity **speeds up**, to the right

The floor pushes up as hard as the weight pulls down, so the normal contact force is 50 N too.

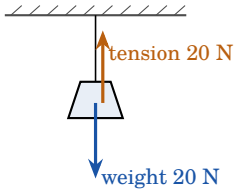


Check it. Every arrow starts at the centre of the object and carries a name and a size. Forces that balance have arrows of the same length.

Task 1 · Draw the free body diagram

Add every force to each object, with its name and size. A missing size? The motion tells you.

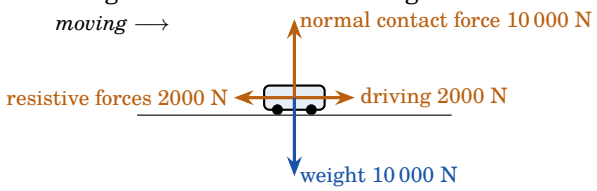
1 A lamp hangs from the ceiling on a cable, not moving. Its weight is 20 N.



Resultant force: **0 N (balanced)**

Velocity: ☑ **stays the same** ☐ speeds up
tick one ☐ slows down ☐ changes direction

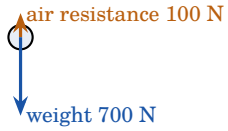
2 A car drives along a flat road at a steady speed. Its driving force is 2000 N. Its weight is 10 000 N.



Resultant force: **0 N (balanced)**

Velocity: ☑ **stays the same** ☐ speeds up
tick one ☐ slows down ☐ changes direction

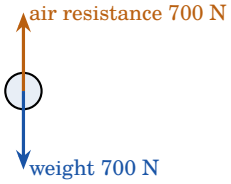
3 A skydiver has just jumped from a plane. Her weight is 700 N. Air resistance on her is 100 N.



Resultant force: **600 N downwards**

Velocity: ☐ stays the same ☑ **speeds up**
tick one ☐ slows down ☐ changes direction

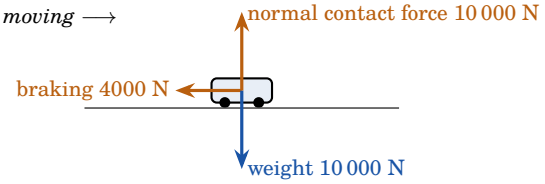
4 Later, the same skydiver falls at a steady speed: her terminal velocity.



Resultant force: **0 N (balanced)**

Velocity: ☑ **stays the same** ☐ speeds up
tick one ☐ slows down ☐ changes direction

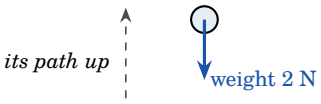
5 A car moving forwards brakes with a braking force of 4000 N. There is no driving force and no air resistance. Weight: 10 000 N.



Resultant force: **4000 N backwards**

Velocity: ☐ stays the same ☐ speeds up
tick one ☑ **slows down** ☐ changes direction

6 A ball thrown straight up is still for an instant at the very top. Its weight is 2 N. There is no air resistance at that instant.



Resultant force: **2 N downwards**

Velocity: ☐ stays the same ☐ speeds up
tick one ☐ slows down ☑ **changes direction**

Task 2 · What do your diagrams tell you?

(a) (i) Which of diagrams 1–6 show an object that is **moving** with a resultant force of **zero**?
(ii) What does that tell you about whether a moving object needs a resultant force?

(i) Diagrams 2 and 4 (the car at a steady speed; the skydiver at terminal velocity). (ii) It does not: with zero resultant force, a moving object keeps moving at the same speed in the same direction. A resultant force *changes* the velocity.

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